Problems In Group Theory

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Contents

Chapter 1

Basic Group Theory

1.1 Introduction

Problem 1.1.1. *Determine which of the following binary operations are associative:*

1. *the operation on* $\mathbb{Z}$ *defined by* $a * b = a - b$
2. *the operation on* $\mathbb{R}$ *defined by* $a * b = a + b + ab$
3. *the operation on* $\mathbb{Q}$ *defined by* $a * b = \frac{a+b}{5}$
4. *the operation on* $\mathbb{Z} \times \mathbb{Z}$ *defined by* $(a, b) * (c, d) = (ad + bc, bd)$
5. *the operation on* $\mathbb{Q} - \{0\}$ *defined by* $a * b = \frac{a}{b}$

Proof. This exercise is pretty trivial as we just have to compute $a * (b * c) = (a * b) * c$. For 1., $a * (b * c) = a * (b - c) = a - (b - c) = a - b + c$ but $(a * b) * c = (a - b) * c = (a - b) - c = a - b - c$. Rest of them are pretty similar. $\square$

Problem 1.1.2. *Prove that addition of residue classes in* $\mathbb{Z}/n\mathbb{Z}$ *is associative.*

Proof. By definition of residue addition,

$$[a] + [b] = [a + b]$$

Therefore, we get the following.

$$\begin{aligned}
[a] + ([b] + [c]) &= [a] + [b + c] \\
&= [a + (b + c)] \\
&= [(a + b) + c] \\
&= [a + b] + [c] \\
&= ([a] + [b]) + [c]
\end{aligned}$$

$\square$

Problem 1.1.3. *Prove that multiplication of residue classes in* $\mathbb{Z}/n\mathbb{Z}$ *is associative.*

Proof. By definition of residue multiplication,

$$[a] \cdot [b] = [a \cdot b]$$

Therefore, we get the following.

$$\begin{aligned} [a] \cdot ([b] \cdot [c]) &= [a] \cdot [b \cdot c] \\ &= [a \cdot (b \cdot c)] \\ &= [(a \cdot b) \cdot c] \\ &= [a \cdot b] \cdot [c] \\ &= ([a] \cdot [b]) \cdot [c] \end{aligned}$$

□

Problem 1.1.4. *Prove for all $n > 1$ that $\mathbb{Z}/n\mathbb{Z}$ is not a group under multiplication of residue classes.*

Proof. There is no multiplicative inverse for $[a]$ with $\gcd(a, n) > 1$. The proof for it pretty trivial, if there was a inverse, it would mean that

$$a \cdot b \equiv 1 \pmod{n} \implies 0 \equiv 1 \pmod{\gcd(a, n)}$$

That's a contradiction. □

Problem 1.1.5. *Determine which of the following sets are groups under addition:*

1. *the set of rational numbers (including $0 = 0/1$) in lowest terms whose denominators are odd*
2. *the set of rational numbers (including $0 = 0/1$) in lowest terms whose denominators are even*
3. *the set of rational numbers of absolute value < 1*
4. *the set of rational numbers of absolute value ≥ 1 together with 0*
5. *the set of rational numbers with denominators equal to 1 or 2*
6. *the set of rational numbers with denominators equal to 1, 2 or 3.*

Proof. Below are the answers:

1. Group.
2. Not a group. Because it is not closed. (Eg: $5/2 + 7/2 = 6$)
3. Not a group. Because again it is not closed. (Eg : $0.99 + 0.98 = 1.97$)
4. Not a group. Because again it is not closed. (Eg : $4 + (-3.1) = 0.9$)
5. Group.
6. Not a group. Because its not closed. (Eg : $1/2 + 1/3 = 5/6$)

□

Problem 1.1.6. *Let $G = \{x \in \mathbb{R} \mid 0 \le x < 1\}$ and for $x, y \in G$ let $x * y$ be the fractional part of $x + y$ (i.e., $x * y = x + y - \lfloor x + y \rfloor$ where $\lfloor a \rfloor$ is the greatest integer less than or equal to a). Prove that $*$ is a well defined binary operation on G and that G is an abelian group under $*$ (called the real numbers mod 1).*

Proof. It is well defined as,

$$x_1 = x_2, y_1 = y_2 \implies x_1 * y_1 = x_1 + y_1 - \lfloor x_1 + y_1 \rfloor$$

$$\implies x_1 + y_1 - \lfloor x_1 + y_1 \rfloor = x_2 + y_2 - \lfloor x_2 + y_2 \rfloor = x_2 * y_2$$

Also, its closed as $x * y = \{x + y\}$ and $0 \le \{x + y\} < 1$. One can check that inverses and identity exists. For associativity,

$$\begin{aligned} x * (y * z) &= x * (y + z - \lfloor y + z \rfloor) \\ &= \{x + y + z - \lfloor y + z \rfloor\} \\ &= \{x + y + z\} \\ &= \{x + y - \lfloor x + y \rfloor + z\} \\ &= (x * y) * z \end{aligned}$$

□

Problem 1.1.7. *Let $G = \{z \in \mathbb{C} \mid z^n = 1 \text{ for some } n \in \mathbb{Z}^+\}$.*

1. *Prove that G is a group under multiplication (called the group of roots of unity in $\mathbb{C}$).*

2. *Prove that G is not a group under addition.*

Proof. For G under multiplication, we know that for all $z_1, z_2, z_3 \in G$ we have

$$z_1 \cdot (z_2 \cdot z_3) = (z_1 \cdot z_2) \cdot z_3$$

Also, it is closed as for all $z_1, z_2 \in G$ we have $z_1^n = 1$ and $z_2^k = 1$ so

$$(z_1 \cdot z_2)^{\text{lcm}(n,k)} = 1$$

Now, for inverse of a $z \in G$ just take $1/z$. As $(1/z)^n = 1/1 = 1$ so $1/z \in G$. Also, $z \cdot 1/z = 1$.
For G under addition, if it was a group then, $1 \in G$ as $1^1 = 1 \in G \implies 1 + 1 = 2 \in G$ but there does not exist any integer n such that $2^n = 1$. □

Problem 1.1.8. *Let $G = \{a + b\sqrt{2} \in \mathbb{R} \mid a, b \in \mathbb{Q}\}$.*

1. *Prove that G is a group under addition.*

2. *Prove that the nonzero elements of G are a group under multiplication.*

Proof. We know reals are associative so the elements of G are associative under addition. The inverse for $a + b\sqrt{2}$ is $(-a) + (-b)\sqrt{2}$. It is also closed as,

$$a + b\sqrt{2} + c + d\sqrt{2} = (a + c) + (b + d)\sqrt{2}$$

The identity is going to be $0 + 0 \cdot \sqrt{2} = 0$.

We know reals are associative so the elements of G are associative under multiplication. The inverse for $a+b\sqrt{2}$ is $\dfrac{a}{a^2-2b^2} - \dfrac{b}{a^2-2b^2}\sqrt{2}$. The identity is $1+0\cdot\sqrt{2}=1$. And its closed under multiplication as

$$(a+b\sqrt{2})(c+d\sqrt{2}) = (ac+2bd)+(ad+bc)\sqrt{2}$$

□

Problem 1.1.9. *Prove that a finite group is abelian if and only if its group table is a symmetric matrix.*

Proof. Let $G=\{g_1,\ldots,g_n\}$ be an group with n elements. Then we define the group matrix to be a $n\times n$ square matrix with entires $a_{ij}=g_ig_j$. So, if its G is abelian then $a_{ij}=g_ig_j=g_jg_i=a_{ji}$, so its symmetric. If the group table is symmetric then $a_{ij}=a_{ji} \implies g_ig_j=g_jg_i$ for all $1\le i,j\le n$. Thus, its abelian. □

Problem 1.1.10. *Find the orders of each element of the additive group $\mathbb{Z}/12\mathbb{Z}$.*

Proof. For $[1],[5],[7],[11]$ its going to be 12. For $[2],[10]$ its going to be 6, for $[4],[8]$ its going to be 3, for $[6]$ its going to be 2. For $[3],[9]$ its going to be 4. For $[0]$ its going to be 1. □

Problem 1.1.11. *Prove that $(a_1a_2\ldots a_n)^{-1}=a_n^{-1}a_{n-1}^{-1}\ldots a_1^{-1}$ for all $a_1,a_2,\ldots,a_n\in G$.*

Proof. We will use strong induction. Its obviously true for $n=1$ and assume its true for $n=2,\ldots,k$. Then,

$$\begin{aligned}((a_1\cdots a_k)\cdot a_{k+1})^{-1} &= a_{k+1}^{-1}(a_1\cdots a_k)^{-1}\\ &= a_{k+1}^{-1}a_k^{-1}\cdots a_1^{-1}\end{aligned}$$

□

Problem 1.1.12. *Let x be an element of G. Prove that $x^2=1$ if and only if $|x|$ is either 1 or 2.*

Proof. Since $x^2=1$ we have $|x|\le 2$, and since $|x|$ is a positive integer we have $|x|=1,2$. And if $|x|=1$ then $x=1 \implies x^2=x=1$ and if $|x|=2$ then $x^2=1$. □

Problem 1.1.13. *Let x be an element of G. Prove that if $|x|=n$ for some positive integer n then $x^{-1}=x^{n-1}$.*

Proof. If $|x|=n$ then $x^n=1 \implies x^{n-1}*x^1=1 \implies x^{n-1}=1*x^{-1}=x^{-1}$. □

Problem 1.1.14. *For x an element in G show that x and x^{-1} have the same order.*

Proof. If $|x|=\infty$ then $|x^{-1}|=\infty$ because if $|x^{-1}|$ where to be finite it would lead to a pretty obvious contradiction. If $|x|=k$ then $x^k=1 \implies x^{-k}=1$ so $|x^{-1}|\le k$ and if $|x^{-1}|=i<k$ then $x^{-i}=1 \implies x^i=1 \implies k\le i<k$ which is a contradiction. □

Problem 1.1.15. *Let G be a finite group and let x be an element of G of order n. Prove that if n is odd, then $x=(x^2)^k$ for some k.*

Proof. Since n is the order of the element x, we have $x^n = 1 \Rightarrow x^{n+1} = x$. Since $n+1 = 2k$ for some $k \in \mathbb{Z}^+$ we have

$$x^{2k} = x \Rightarrow (x^2)^k = x$$

□

Problem 1.1.16. *If x and g are elements of the group G, prove that $|x| = |g^{-1}xg|$. Deduce that $|ab| = |ba|$ for all $a, b \in G$.*

Proof. We can easily prove through induction that $(g^{-1}xg)^k = g^{-1}x^kg$. Now let $|x| = n$ and let $|g^{-1}xg| = m$ and suppose $m < n$ then

$$g^{-1}x^mg = e \Rightarrow x^m = e \Rightarrow |x| = n \leq m < n$$

Thus, $m \geq n$ and $g^{-1}x^ng = g^{-1}eg = e$ so we have $|g^{-1}xg| = n = |x|$.
Also, if you let $x = ab$ and $g = a$ we get,

$$\begin{aligned} |ab| &= |(a)^{-1}(ab)(a)| \\ &= |a^{-1}(ab)a| \\ &= |ba| \end{aligned}$$

□

Problem 1.1.17. *Suppose $x \in G$ and $|x| = n < \infty$. If $n = st$ for some positive integers s and t, prove that $|x^s| = t$.*

Proof. If $|x| = n$ and $n = st$ then $x^n = 1 \Rightarrow x^{st} = 1 \Rightarrow (x^s)^t = 1$. Thus, $|x^s| \leq t$. Suppose $|x^s| = k < t$ then $x^{sk} = 1$ but $n = st \leq sk \Rightarrow t \leq k < t$. Thus, $|x^s| = t$. □

Problem 1.1.18. *Prove that if $x^2 = 1$ for all $x \in G$ then G is abelian.*

Proof. We need to prove that $xy = yx$ for all $y, x \in G$. We know that $x^2 = y^2 = 1$ so

$$xy = x^{-1}y^{-1} = (yx)^{-1} = yx$$

as $(xy)^2 = 1$. □

Problem 1.1.19. *Prove that if x is an element of the group G then $C = \{x^n \mid n \in \mathbb{Z}\}$ is a subgroup of G (called the cyclic subgroup of G generated by x).*

Proof. We know the elements of C are associative because they're the element of G. The identity is going to be $x^0 = 1$ and the inverse of x^n is x^{-n} as

$$x^n \cdot x^{-n} = x^{n-n} = 1$$

Also it's closed as $x^k \cdot x^\ell = x^{k+\ell} \in C$. □

Problem 1.1.20. *Let $(A, *)$ and $(B, \circ)$ be groups and let $A \times B$ be their direct product. Verify all the group axioms for $A \times B$:*

1. *Prove that the associative law holds: for all $(a_i, b_i) \in A \times B, i = 1, 2, 3$,*
   $(a_1, b_1)[(a_2, b_2)(a_3, b_3)] = [(a_1, b_1)(a_2, b_2)](a_3, b_3)$,

2. *Prove that* $(1,1)$ *is the identity of* $A \times B$*, and*

3. *Prove that the inverse of* (a,b) *is* (a^{-1}, b^{-1}).

Proof. It's just the matter of calculation. □

Problem 1.1.21. *Prove that* $A \times B$ *is an abelian group if and only if both* A *and* B *are abelian.*

Proof. Suppose $A \times B$ is abelian then,

$$\begin{aligned}(a,b)(c,d) &= (c,d)(a,b)\\ (a*c, b\circ d) &= (c*a, d\circ b)\end{aligned}$$

Thus $a * c = c * a$ and $b \circ d = d \circ b$. Therefore, both A and B are abelian.
And if A and B are abelian then

$$\begin{aligned}(a,b)(c,d) &= (a*c, b\circ d)\\ &= (c*a, d\circ b)\\ &= (c,d)(a,b)\end{aligned}$$

□

Problem 1.1.22. *Suppose* G *is a group and* $|x| = n < \infty$*. If* $x^a = 1$ *then* $n \mid a$.

Proof. Using the Euclidian algorithm we get,

$$a = nq + r$$

where $0 \le r < n$ Then,

$$x^a = x^{nq+r} = x^r = 1$$

But that's a contradiction unless $r = 0$. □

Problem 1.1.23. *Prove that the elements* $(a,1)$ *and* $(1,b)$ *of* $A \times B$ *commute and deduce that the order of* (a,b) *is the least common multiple of* $|a|$ *and* $|b|$.

Proof. For the first statement,

$$(a,1)(1,b) = (a*1, 1\circ b) = (1*a, b\circ 1) = (1,b)(a,1)$$

Now, suppose $|(a,b)| = n$ then,

$$\begin{aligned}(a,b)^n &= (1,1)\\ \implies (a^n, b^n) &= (1,1)\end{aligned}$$

Thus, $a^n = 1$ and $b^n = 1$. Therefore, $n \mid |a|$ and $n \mid |b|$. Thus, $n \mid \text{lcm}(|a|,|b|)$ which implies that $n \ge \text{lcm}(|a|,|b|)$. Now, $n = \text{lcm}(|a|,|b|)$ satisfies $(a,b)^n = (1,1)$. □

Problem 1.1.24. *Prove that any finite group* G *of even order contains an element of order 2.*

Proof. Let $\iota(G) = \{g \in G \mid g \ne g^{-1}\}$. Then notice that order of $\iota(G)$ is even as it contains both g, g^{-1}. Then, the order of $G - \iota(G)$ is at least 2 because the order of G is even. Thus, we have a non identity element $x \in G - \iota(G)$ such that $x^2 = 1$. □

Problem 1.1.25. *If x is an element of finite order n in G, prove that the elements of*

$$C = \{x^k \mid 0 \leq k \leq n-1\}$$

are all distinct. Deduce that $|x| \leq |G|$.

Proof. Suppose that for $a \neq b$ we have,

$$x^a = x^b$$

Then, $x^{a-b} = 1$ which implies $n \mid a-b$ but $a-b < |a| < n$ which is a contradiction. Thus, all the elements of C are distinct. And since C is a subgroup we have

$$|x| = n = |C| \leq |G|$$

□

Problem 1.1.26. *Let x be an element of finite order n in G.*

1. *Prove that if n is odd then $x^i \neq x^{-i}$ for all $i = 1, 2, \ldots, n-1$.*
2. *Prove that if $n = 2k$ and $1 \leq i < n$ then $x^i = x^{-i}$ if and only if $i = k$.*

Proof. If n is odd and $x^i = x^{-i}$ then $x^{2i} = 1$ which implies $n \mid 2i \implies n \mid i$ but thats not possible as $i < n$. If $n = 2k$ and $x^i = x^{-i} \Rightarrow x^{2i} = 1$ which implies $2k \mid 2i$ and since $2 \leq 2i < 2n$ we have $2i = n$. □

Problem 1.1.27. *Assume $G = \{1, a, b, c\}$ is a group of order 4 with identity 1. Assume also that G has no elements of order 4. Prove that there is a unique group table for G. Deduce that G is abelian.*

Proof. Notice that $a^2 = b^2 = c^2 = 1$, as $|a|$ divides $|G| = 4$ so $|a| = 2$. If it were to be 1 it would mean that $a = 1$ and it can't be 4. Now $ab = c$ as $ab = 1, a, b$ would all lead to trivial contradiction. Doing mere simple calculation we can find out the group table for this group which is,

$\cdot$	1	a	b	c
1	1	a	b	c
a	a	1	c	b
b	b	c	1	a
c	c	b	a	1

Since the table is symmetric its abelian. □

1.2 Diheral Group